

Creation Date 06-Apr-2021

Revision Date 06-Apr-2021

Revision Number 1

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: **n-Heptane: Acetone: 90:10 % v/v**
Cat No. : **SP/3967/27; SP/3967/15**

Unique Formula Identifier (UFI) **AH6X-X6QP-4X0J-DA96**

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Company

EU entity/business name
Acros Organics BVBA
Janssen Pharmaceuticaaan 3a
2440 Geel, Belgium
General Info; Tel: +32-14-57 52 11
(info@acros.com)
Technical Support; Tel +32-14-56 56 00
(acros.techsupport@thermofisher.com)

UK entity/business name
Fisher Scientific UK
Bishop Meadow Road, Loughborough,
Leicestershire LE11 5RG, United Kingdom
General info; Tel: +44 (0)1509 231166

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
e-mail - infoch@thermofisher.com

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

Tel: 01509 231166
Chemtrec US: (800) 424-9300
Chemtrec EU: 001-703-527-3887

For customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

**Poison Centre - Emergency
information services**

Ireland : National Poisons Information Centre (NPIC) -
01 809 2166 (8am-10pm, 7 days a week)
Malta : +356 2395 2000
Cyprus : +357 2240 5611

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Flammable liquids

Category 2 (H225)

Health hazards

Aspiration Toxicity

Category 1 (H304)

Skin Corrosion/Irritation

Category 2 (H315)

Serious Eye Damage/Eye Irritation

Category 2 (H319)

Specific target organ toxicity - (single exposure)

Category 3 (H336)

Environmental hazards

Acute aquatic toxicity

Category 1 (H400)

Chronic aquatic toxicity

Category 1 (H410)

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H225 - Highly flammable liquid and vapor

H304 - May be fatal if swallowed and enters airways

H315 - Causes skin irritation

H319 - Causes serious eye irritation

H336 - May cause drowsiness or dizziness

H410 - Very toxic to aquatic life with long lasting effects

EUH066 - Repeated exposure may cause skin dryness or cracking

Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P280 - Wear protective gloves/protective clothing/eye protection/face protection

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower
P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician
P331 - Do NOT induce vomiting
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P337 + P313 - If eye irritation persists: Get medical advice/attention

2.3. Other hazards

No information available

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
n-Heptane	142-82-5	EEC No. 205-563-8	88-90	Flam. Liq. 2 (H225) Asp. Tox. 1 (H304) Skin Irrit. 2 (H315) STOT SE 3 (H336) Aquatic Acute 1 (H400) Aquatic Chronic 1 (H410)
Acetone	67-64-1	EEC No. 200-662-2	10-12	Flam. Liq. 2 (H225) Eye Irrit. 2 (H319) STOT SE 3 (H336) EUH066

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
n-Heptane	-	1	-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	If symptoms persist, call a physician.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
Ingestion	Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically. Symptoms may be delayed.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Water mist may be used to cool closed containers.

Extinguishing media which must not be used for safety reasons

Do not use a solid water stream as it may scatter and spread fire.

5.2. Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Do not allow run-off from fire-fighting to enter drains or water courses.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂).

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Take precautionary measures against static discharges. Use spark-proof tools and explosion-proof equipment.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Class 3

Switzerland - Storage of hazardous substances

Storage class - SC 3

<https://www.kvu.ch/de/themen/stoffe-und-produkte>

<https://www.kvu.ch/fr/themes/substances-et-produits>

<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

Component	European Union	The United Kingdom	France	Belgium	Spain
n-Heptane	TWA: 500 ppm (8h) TWA: 2085 mg/m ³ (8h)	STEL: 1500 ppm 15 min STEL: 6255 mg/m ³ 15 min TWA: 500 ppm 8 hr TWA: 2085 mg/m ³ 8 hr	TWA / VME: 400 ppm (8 heures). restrictive limit TWA / VME: 1668 mg/m ³ (8 heures). restrictive limit TWA / VME: 1000 mg/m ³ (8 heures). restrictive limit STEL / VLCT: 500 ppm. restrictive limit STEL / VLCT: 2085 mg/m ³ . restrictive limit STEL / VLCT: 1500 mg/m ³ .	TWA: 400 ppm 8 uren TWA: 1664 mg/m ³ 8 uren STEL: 500 ppm 15 minuten STEL: 2085 mg/m ³ 15 minuten	TWA / VLA-ED: 500 ppm (8 horas) TWA / VLA-ED: 2085 mg/m ³ (8 horas)
Acetone	TWA: 500 ppm (8h) TWA: 1210 mg/m ³ (8h)	TWA: 500 ppm TWA: 1210 mg/m ³ STEL: 1500 ppm STEL: 3620 mg/m ³	TWA / VME: 500 ppm (8 heures). restrictive limit TWA / VME: 1210 mg/m ³ (8 heures). restrictive limit STEL / VLCT: 1000 ppm. restrictive limit STEL / VLCT: 2420 mg/m ³ . restrictive limit	TWA: 500 ppm 8 uren TWA: 1210 mg/m ³ 8 uren STEL: 1000 ppm 15 minuten STEL: 2420 mg/m ³ 15 minuten	TWA / VLA-ED: 500 ppm (8 horas) TWA / VLA-ED: 1210 mg/m ³ (8 horas)

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

Component	Italy	Germany	Portugal	The Netherlands	Finland
n-Heptane	TWA: 500 ppm 8 ore. Media Ponderata nel Tempo TWA: 2085 mg/m ³ 8 ore. Media Ponderata nel Tempo	TWA: 500 ppm (8 Stunden). AGW - exposure factor 1 TWA: 2100 mg/m ³ (8 Stunden). AGW - exposure factor 1 TWA: 500 ppm (8 Stunden). MAK TWA: 2100 mg/m ³ (8 Stunden). MAK Höhepunkt: 500 ppm Höhepunkt: 2100 mg/m ³	STEL: 500 ppm 15 minutos TWA: 500 ppm 8 horas TWA: 2085 mg/m ³ 8 horas	STEL: 1600 mg/m ³ 15 minuten TWA: 1200 mg/m ³ 8 uren	TWA: 300 ppm 8 tunteina TWA: 1200 mg/m ³ 8 tunteina STEL: 500 ppm 15 minuutteina STEL: 2100 mg/m ³ 15 minuutteina
Acetone	TWA: 500 ppm 8 ore. Media Ponderata nel Tempo TWA: 1210 mg/m ³ 8 ore. Media Ponderata nel Tempo	TWA: 500 ppm TWA: 1200 mg/m ³	STEL: 750 ppm 15 minutos TWA: 500 ppm 8 horas TWA: 1210 mg/m ³ 8 horas	STEL: 2420 mg/m ³ 15 minuten TWA: 1210 mg/m ³ 8 uren	TWA: 500 ppm 8 tunteina TWA: 1200 mg/m ³ 8 tunteina STEL: 630 ppm 15 minuutteina STEL: 1500 mg/m ³ 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
n-Heptane	MAK-KZGW: 2000 ppm 15 Minuten MAK-KZGW: 8000 mg/m ³ 15 Minuten MAK-TMW: 500 ppm 8 Stunden MAK-TMW: 2000 mg/m ³ 8 Stunden	TWA: 200 ppm 8 timer TWA: 820 mg/m ³ 8 timer	STEL: 400 ppm 15 Minuten STEL: 1600 mg/m ³ 15 Minuten TWA: 400 ppm 8 Stunden TWA: 1600 mg/m ³ 8 Stunden	STEL: 2000 mg/m ³ 15 minutach TWA: 1200 mg/m ³ 8 godzinach	TWA: 200 ppm 8 timer TWA: 800 mg/m ³ 8 timer TWA: 40 ppm 8 timer TWA: 275 mg/m ³ 8 timer STEL: 250 ppm 15 minutter. value calculated STEL: 1000 mg/m ³ 15 minutter. value calculated
Acetone	MAK-KZGW: 2000 ppm 15 Minuten MAK-KZGW: 4800 mg/m ³ 15 Minuten MAK-TMW: 500 ppm 8 Stunden MAK-TMW: 1200 mg/m ³ 8 Stunden	TWA: 250 ppm 8 timer TWA: 600 mg/m ³ 8 timer	STEL: 1000 ppm 15 Minuten STEL: 2400 mg/m ³ 15 Minuten TWA: 500 ppm 8 Stunden TWA: 1200 mg/m ³ 8 Stunden	STEL: 1800 mg/m ³ 15 minutach TWA: 600 mg/m ³ 8 godzinach	TWA: 125 ppm 8 timer TWA: 295 mg/m ³ 8 timer STEL: 156.25 ppm 15 minutter. value calculated STEL: 368.75 mg/m ³ 15 minutter. value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
n-Heptane	TWA: 1600 mg/m ³	kože TWA-GVI: 500 ppm 8 satima. TWA-GVI: 2085 mg/m ³ 8 satima.	TWA: 500 ppm 8 hr. TWA: 2085 mg/m ³ 8 hr. STEL: 1500 ppm 15 min STEL: 6255 mg/m ³ 15 min	TWA: 500 ppm TWA: 2085 mg/m ³	TWA: 1000 mg/m ³ 8 hodinách. Ceiling: 2000 mg/m ³
Acetone	TWA: 600 mg/m ³ STEL: 1400 mg/m ³	TWA-GVI: 500 ppm 8 satima. TWA-GVI: 1210 mg/m ³ 8 satima.	TWA: 500 ppm 8 hr. TWA: 1210 mg/m ³ 8 hr. STEL: 1500 ppm 15 min STEL: 3630 mg/m ³ 15 min	Skin-potential for cutaneous absorption TWA: 500 ppm TWA: 1210 mg/m ³	TWA: 800 mg/m ³ 8 hodinách. Ceiling: 1500 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
n-Heptane	TWA: 500 ppm 8 tundides. TWA: 2085 mg/m ³ 8 tundides.	TWA: 500 ppm 8 hr TWA: 2085 mg/m ³ 8 hr	STEL: 500 ppm STEL: 2000 mg/m ³ TWA: 500 ppm TWA: 2000 mg/m ³	TWA: 2000 mg/m ³ 8 órában. AK	TWA: 200 ppm 8 klukkustundum. TWA: 820 mg/m ³ 8 klukkustundum. Ceiling: 400 ppm Ceiling: 1640 mg/m ³
Acetone	TWA: 500 ppm 8 tundides. TWA: 1210 mg/m ³ 8 tundides.	TWA: 500 ppm 8 hr TWA: 1210 mg/m ³ 8 hr	STEL: 3560 mg/m ³ TWA: 1780 mg/m ³	TWA: 1210 mg/m ³ 8 órában. AK	TWA: 250 ppm 8 klukkustundum. TWA: 600 mg/m ³ 8 klukkustundum. Ceiling: 500 ppm Ceiling: 1200 mg/m ³

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
n-Heptane	STEL: 500 ppm STEL: 2085 mg/m ³ TWA: 85 ppm TWA: 350 mg/m ³	TWA: 500 ppm IPRD TWA: 2085 mg/m ³ IPRD STEL: 750 ppm STEL: 3128 mg/m ³	TWA: 500 ppm 8 Stunden TWA: 2085 mg/m ³ 8 Stunden	TWA: 500 ppm TWA: 2085 mg/m ³	TWA: 500 ppm 8 ore TWA: 2085 mg/m ³ 8 ore
Acetone	TWA: 500 ppm TWA: 1210 mg/m ³	TWA: 500 ppm IPRD TWA: 1210 mg/m ³ IPRD STEL: 1000 ppm STEL: 2420 mg/m ³	TWA: 500 ppm 8 Stunden TWA: 1210 mg/m ³ 8 Stunden	TWA: 500 ppm TWA: 1210 mg/m ³	TWA: 500 ppm 8 ore TWA: 1210 mg/m ³ 8 ore

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
n-Heptane		TWA: 500 ppm TWA: 2085 mg/m ³	TWA: 500 ppm 8 urah applies to all isomers TWA: 2085 mg/m ³ 8 urah applies to all isomers STEL: 500 ppm 15 minutah applies to all isomers STEL: 2085 mg/m ³ 15 minutah applies to all isomers	Indicative STEL: 300 ppm 15 minuter Indicative STEL: 1200 mg/m ³ 15 minuter TLV: 200 ppm 8 timmar. NGV TLV: 800 mg/m ³ 8 timmar. NGV	TWA: 500 ppm 8 saat TWA: 2085 mg/m ³ 8 saat
Acetone	TWA: 200 mg/m ³ 1763 MAC: 800 mg/m ³	TWA: 500 ppm TWA: 1210 mg/m ³	TWA: 500 ppm 8 urah TWA: 1210 mg/m ³ 8 urah STEL: 2420 mg/m ³ 15 minutah STEL: 1000 ppm 15 minutah	Indicative STEL: 500 ppm 15 minuter Indicative STEL: 1200 mg/m ³ 15 minuter TLV: 250 ppm 8 timmar. NGV TLV: 600 mg/m ³ 8 timmar. NGV	TWA: 500 ppm 8 saat TWA: 1210 mg/m ³ 8 saat

Biological limit values

List source(s):

Component	European Union	United Kingdom	France	Spain	Germany
Acetone			Acetone: 100 mg/L urine end of shift	Acetone: 50 mg/L urine end of shift	Acetone: 80 mg/L urine (end of shift)

Component	Italy	Finland	Denmark	Bulgaria	Romania
Acetone				Acetone: 80 mg/L urine at the end of exposure or end of work shift	Acetone: 50 mg/L urine end of shift

Component	Gibraltar	Latvia	Slovak Republic	Luxembourg	Turkey
Acetone			Acetone: 80 mg/L urine end of exposure or work shift		

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
n-Heptane 142-82-5 (88-90)				DNEL = 300mg/kg bw/day
Acetone 67-64-1 (10-12)				DNEL = 186mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
n-Heptane 142-82-5 (88-90)				DNEL = 2085mg/m ³
Acetone 67-64-1 (10-12)	DNEL = 2420mg/m ³			DNEL = 1210mg/m ³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Acetone 67-64-1 (10-12)	PNEC = 10.6mg/L	PNEC = 30.4mg/kg sediment dw	PNEC = 21mg/L	PNEC = 100mg/L	PNEC = 29.5mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Acetone 67-64-1 (10-12)	PNEC = 1.06mg/L	PNEC = 3.04mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles (European standard - EN 166)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber	See manufacturers recommendations	-	EN 374	(minimum requirement)
Viton (R)				

Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: low boiling organic solvent Type AX Brown conforming to

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387

Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance		
Odor	No information available	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	Highly flammable	Estimated
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	-20 °C / -4 °F	Method - Estimated
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	Not applicable	
Viscosity	No data available	
Water Solubility	Immiscible	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
n-Heptane	4.66	
Acetone	-0.24	
Vapor Pressure	No data available	
Density / Specific Gravity	0.69	
Bulk Density	Not applicable	Liquid
Vapor Density	No data available	(Air = 1.0)
Particle characteristics	No data available	

9.2. Other information

VOC Content(%) 100 Vapors may form explosive mixtures with air

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

Hazardous Polymerization No information available.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition.

10.5. Incompatible materials

Strong oxidizing agents.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO₂).

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
n-Heptane	>2000 mg/kg (rat)	LD50 = 3000 mg/kg (Rabbit)	LC50 > 73.5 mg/L (Rat) 4 h
Acetone	5800 mg/kg (Rat)	> 15800 mg/kg (rabbit) > 7400 mg/kg (rat)	76 mg/l, 4 h, (rat)

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

Component	Test method	Test species	Study result
Acetone 67-64-1 (10-12)	Guinea Pig Maximisation Test (GPMT)	guinea pig	non-sensitising

(e) germ cell mutagenicity; No data available

Component	Test method	Test species	Study result
Acetone 67-64-1 (10-12)	OECD Test Guideline 471 AMES test	in vivo	negative
	OECD Test Guideline 476 Mammalian Gene cell mutation	in vitro	negative

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

(g) reproductive toxicity; No data available

(h) STOT-single exposure; Category 3

Results / Target organs Central nervous system (CNS).

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; Category 1

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
n-Heptane	LC50: = 375.0 mg/L, 96h (Cichlid fish)	EC50: >10 mg/L/24h	
Acetone	Oncorhynchus mykiss: LC50 = 5540 mg/l 96h Alburnus alburnus: LC50 = 11000 mg/l 96h Leuciscus idus: LC50 = 11300 mg/L/48h Salmo gairdneri: LC50 = 6100 mg/L/24h	EC50 = 8800 mg/L/48h EC50 = 12700 mg/L/48h EC50 = 12600 mg/L/48h	NOEC = 430 mg/l (algae; 96 h)

Component	Microtox	M-Factor
n-Heptane		1
Acetone	EC50 = 14500 mg/L/15 min	

12.2. Persistence and degradability

Persistence May persist.

Component	Degradability
Acetone 67-64-1 (10-12)	91 % (28 d) (OECD 301 B)

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential

Product has a high potential to bioconcentrate

Component	log Pow	Bioconcentration factor (BCF)
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SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

n-Heptane	4.66	No data available
Acetone	-0.24	0.69

12.4. Mobility in soil

Spillage unlikely to penetrate soil The product is insoluble and floats on water Is not likely mobile in the environment due its low water solubility. Is not likely mobile in the environment due its low water solubility and propensity to bind to soil particles

12.5. Results of PBT and vPvB assessment

No data available for assessment.

12.6. Endocrine disrupting properties

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number

UN1993

14.2. UN proper shipping name

FLAMMABLE LIQUID, N.O.S.
(contains n-HEPTANE and ACETONE)

14.3. Transport hazard class(es)

3

14.4. Packing group

II

ADR

14.1. UN number

UN1993

14.2. UN proper shipping name

FLAMMABLE LIQUID, N.O.S.

FSUSP3967

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

Technical Shipping Name	(contains n-HEPTANE and ACETONE)
14.3. Transport hazard class(es)	3
14.4. Packing group	II

IATA

14.1. UN number	UN1993
14.2. UN proper shipping name	FLAMMABLE LIQUID, N.O.S.
Technical Shipping Name	(contains n-HEPTANE and ACETONE)
14.3. Transport hazard class(es)	3
14.4. Packing group	II

14.5. Environmental hazards	Dangerous for the environment Product is a marine pollutant according to the criteria set by IMDG/IMO
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14.6. Special precautions for user	No special precautions required
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14.7. Maritime transport in bulk according to IMO instruments	Not applicable, packaged goods
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SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
n-Heptane	142-82-5	205-563-8	-	-	X	X	KE-18271	X	X
Acetone	67-64-1	200-662-2	-	-	X	X	KE-29367	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDL	AICS	NZIoC	PICCS
n-Heptane	142-82-5	X	ACTIVE	X	-	X	X	X
Acetone	67-64-1	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Component	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
n-Heptane	-	Use restricted. See item 75. (see link for restriction details)	-
Acetone	-	Use restricted. See item 75. (see link for restriction details)	-

<https://echa.europa.eu/substances-restricted-under-reach>

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
n-Heptane	142-82-5	Not applicable	Not applicable
Acetone	67-64-1	Not applicable	Not applicable

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification Water endangering class = 2 (self classification)

Component	Germany - Water Classification (VwVws)	Germany - TA-Luft Class
n-Heptane	WGK2	
Acetone	WGK1	

Component	France - INRS (Tables of occupational diseases)
n-Heptane	Tableaux des maladies professionnelles (TMP) - RG 84
Acetone	Tableaux des maladies professionnelles (TMP) - RG 84

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
n-Heptane 142-82-5 (88-90)	Prohibited and Restricted Substances	Group I	
Acetone 67-64-1 (10-12)		Group I	

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

H304 - May be fatal if swallowed and enters airways

H315 - Causes skin irritation

H319 - Causes serious eye irritation

H336 - May cause drowsiness or dizziness

H400 - Very toxic to aquatic life

H410 - Very toxic to aquatic life with long lasting effects

EUH066 - Repeated exposure may cause skin dryness or cracking

H225 - Highly flammable liquid and vapor

Legend

CAS - Chemical Abstracts Service

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

SAFETY DATA SHEET

n-Heptane: Acetone: 90:10 % v/v

Revision Date 06-Apr-2021

PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (volatile organic compound)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>
Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

Creation Date	06-Apr-2021
Revision Date	06-Apr-2021
Revision Summary	Initial Release.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and
Preparations).**

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet